Thesis Defense — Questions & Answers

AI-Based Cyclone Early Warning System

Prepared for Master's Defense | 2026 | 8 Categories · 25+ Questions

This document covers every likely question from thesis committee members, with nested follow-up questions and full answers. Read the main answer first; follow-up questions are indented below each main question.

1. Project Overview

Q1: What is the core research question of this project?

Will adding environmental (ERA5) and satellite (GridSat-B1) data to a machine learning model significantly improve tropical cyclone intensification prediction in the Bay of Bengal compared to using track history alone?

■ Why the Bay of Bengal specifically?

The BoB accounts for ~7% of global cyclone activity but causes disproportionate mortality due to densely populated, low-lying coasts in Bangladesh, India, and Myanmar. Improved early warning directly saves lives.

■ What does 'intensification' mean here?

A significant increase of ≥15 knots in maximum sustained wind speed within a specified forecast window. This threshold was chosen as a balance between scientific relevance and label frequency; the literature standard for Rapid Intensification is ≥30 kt/24h.

Q2: What are the three prediction components?

(1) Intensification classifier: binary — will wind increase ≥15 kt at 48–72h or 120–168h? (2) Track forecaster: where will the cyclone be in the next 48h (8 waypoints × 6h)? (3) Storm surge estimator: how high will surge reach at 8 coastal stations?

■ Which is the most important for early warning?

Intensification — because it determines whether evacuation orders need to be upgraded. Track provides direction context; surge translates that into coastal impact.

Q3: Why three versions of the intensification model?

To demonstrate that each additional data modality improves skill. V1 (track only) establishes a baseline; V2 (+ ERA5) shows the impact of environmental context; V3 (+ satellite IR) shows the additional benefit of spatial imagery. This progression is the thesis's central narrative.

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2. Data & Preprocessing

Q1: Why use IBTrACS instead of JTWC or IMD data directly?

IBTrACS harmonises data from all Regional Specialised Meteorological Centres into a single format with consistent metadata. WMO_WIND provides a standardised reference wind across agencies. For reproducibility, a single public dataset is preferable to agency-specific files.

■ What is the IBTrACS wind source used here?

WMO_WIND — the World Meteorological Organisation consensus wind speed in knots, 1-minute average. USA_WIND (10-minute) was not used as it requires a conversion factor.

Q2: What ERA5 variables were used and why?

SST (sea surface temperature) — primary energy source for cyclone intensification. Mean sea-level pressure — determines storm strength. U10/V10 — surface wind field. T2M/D2M — atmospheric thermodynamic state. Total precipitation — moisture availability. These are the most computationally accessible proxies for the SHIPS environmental indices.

■ What is the difference between ERA5 reanalysis and a real-time forecast?

ERA5 is a retrospective analysis: all available observations are assimilated, quality-controlled, and interpolated onto a uniform grid. It represents the best estimate of past atmospheric state. A real-time forecast only uses data available at the time of issue. This is why the dashboard uses Open-Meteo as a live proxy — the models were trained on ERA5 but served on real-time data (distribution shift).

■ How was ERA5 matched to IBTrACS storm positions?

Using xarray's nearest-neighbour selection: for each track observation, the ERA5 grid cell closest in lat/lon and the ERA5 time step closest in ISO_TIME were selected.

Q3: How were the training/test sets split?

Time-based split: 2018–2019 storms for training, 2020 storms for testing. GroupShuffleSplit by storm SID ensures no rows from the same storm appear in both sets, preventing leakage from sequential track observations. Random row-level splits are inappropriate for time-series data.

■ What is data leakage and why does it matter?

Leakage occurs when the model sees information during training that would not be available at prediction time. For sequential track data, a random split could put hour t+6 in training and hour t in test, letting the model 'see the future'. This inflates evaluation metrics without real predictive improvement.

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3. Machine Learning Methodology

Q1: Why XGBoost for V1 and V2?

XGBoost handles mixed tabular features well, is robust to outliers, provides built-in feature importance, trains quickly on small datasets (<10k rows), and has strong performance on structured prediction tasks per published benchmarks. It also supports scale_pos_weight for class imbalance correction natively.

■ Did you try other models?

XGBoost was selected after considering logistic regression (too linear for the non-linear interaction between SST and wind shear) and Random Forest (similar performance but slower inference). A future ablation could formalise this comparison.

Q2: Why is V1 long-range AUC so low (0.10)?

Three reasons: (1) class imbalance — very few storms in 2018–2020 intensified at 120–168h lead time; (2) track features alone carry almost no signal about environmental support at 5–7 days; (3) the small test set (2020 only, 6 storms) makes the metric volatile. Adding ERA5 (V2) fixed issues 1 and 2, raising AUC to 0.80.

■ Isn't 0.10 worse than random?

AUC = 0.5 is random; 0.10 means the model was systematically inverting its predictions — a sign of severe class imbalance and poor feature-label alignment. This is documented as an expected baseline failure, motivating V2.

Q3: How does the CNN-LSTM (V3) work?

A TimeDistributed CNN encodes each satellite IR patch (64×64 px, 3 conv layers) into a 128-d embedding per time step. An LSTM processes the sequence of up to 8 embeddings (24h window) to capture temporal evolution of the storm's cloud structure. A separate MLP branch encodes tabular ERA5+track features (64-d). Both branches are concatenated and passed through Dense(256) → Dense(1, sigmoid). Trained with BCELoss and Adam for 30 epochs.

■ Why not use a pretrained CNN like ResNet?

445 training images is too few for fine-tuning a large pretrained model reliably without extensive regularisation. The small custom CNN was designed for this data scale. A future improvement would use transfer learning from tropical meteorology image datasets.

■ Why LSTM instead of Transformer attention?

LSTM was chosen for simplicity given the short sequence length (≤8 frames). Temporal attention would be a valid improvement — it can focus on the specific frames showing eyewall organisation or rapid cloud-top cooling.

Q4: How does the track predictor work?

A MultiOutputRegressor (Random Forest, 300 trees, max_depth=10) takes a 26-feature input vector built from the last 4 observations (24h history): per-step lat, lon, wind, and their deltas, plus month and hour. It predicts 16 values — 8 pairs of (dlat, dlon) representing cumulative displacements from the current position to each future waypoint. Reconstruction: predicted_position[k] = current_position + pred[k]. MAE was evaluated with leave-one-storm-out GroupKFold to avoid within-storm leakage.

■ Why not a sequential model like LSTM for track prediction?

Random Forest with multi-output regression is a strong baseline for short-horizon track prediction and trains in minutes on the available data. LSTM would be preferred with larger datasets or longer horizons (>72h).

■ What are the honest accuracy numbers?

24h MAE ~18 km and 48h MAE ~134 km (leave-one-storm-out CV). Earlier in-sample estimates (30/61 km) were inflated by training-data leakage and have been corrected. The 48h degradation is primarily due to missing 500 hPa steering wind features.

■ What is the difference between cumulative displacement and step-to-step delta targets?

Cumulative: pred[k] = lat[t+k] - lat[t]. Step-to-step: pred[k] = lat[t+k] - lat[t+k-1]. This model uses cumulative displacement, so each output is independent and can be directly added to the current position without accumulating step errors.

4. Evaluation & Results

Q1: What metrics are used and why?

ROC-AUC: measures discrimination ability, insensitive to threshold and class balance — standard for binary classifiers with imbalanced classes. PR-AUC: precision-recall area under curve, more informative than ROC when positives are rare (relevant for long-range intensification). Brier Score: measures probabilistic calibration — important because the dashboard displays '% chance' and users deserve reliable probabilities. MAE (km) for track: standard track verification metric in operational meteorology.

■ What is Brier Score?

Brier Score = mean((predicted_probability - actual_label)^2). Range 0–1; lower is better. A score of 0.25 equals always predicting 0.5. It measures both calibration (do predicted probabilities match observed frequencies?) and resolution (does the model distinguish events from non-events?).

Q2: How do you interpret ROC-AUC 0.89 for V3?

AUC 0.89 means that in 89% of randomly chosen (positive, negative) pairs, the model ranks the positive (intensifying storm) higher. This is a strong result for a multi-modal prototype with 445 training images. However, the model has not been validated on out-of-sample storms from other years or basins, so generalisation is unproven.

Q3: Why is V3 not integrated into the live dashboard?

V3 requires PyTorch at runtime and loads a large model file. The dashboard currently uses lightweight joblib-serialised XGBoost models (v2_short.pkl = 329 KB, v2_long.pkl = 248 KB). V3 export to TorchScript or ONNX would enable deployment without a PyTorch dependency — this is listed as future work.

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5. Dashboard & Deployment

Q1: How does the live dashboard work?

On each load: (1) GDACS RSS feed is parsed for active BoB cyclone alerts; (2) Open-Meteo marine + atmospheric API is queried for SST, pressure, wind; (3) If a cyclone is found AND weather is available, V2 XGBoost predicts intensification probability; (4) Track predictor forecasts 48h positions; (5) Surge heuristic estimates coastal risk; (6) Folium map renders all layers. If no active cyclone is found, the dashboard shows standby mode.

■ What is the GDACS feed?

GDACS (Global Disaster Alert and Coordination System) is a European Commission service that aggregates alerts from WMO, NOAA, and regional agencies. The RSS feed provides near-real-time cyclone position, category, and wind speed.

■ What if the GDACS feed is down?

The dashboard falls back to manual input mode, allowing the user to enter storm coordinates and parameters directly for scenario analysis.

Q2: What is the distribution shift problem between training and serving?

V2 was trained on ERA5 reanalysis — a retrospective, quality-controlled, gridded dataset with consistent variable definitions. In production, it is served with Open-Meteo real-time API data. Differences include: spatial resolution (ERA5 0.25° vs Open-Meteo point), temporal averaging windows, QC procedures, and SST retrieval methods. This mismatch means the model may be systematically miscalibrated in production, which is documented as a limitation.

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6. Storm Surge Risk

Q1: How is storm surge risk calculated?

A physics-based heuristic combining: (1) Inverse barometer: ~1 cm per hPa below 1013 hPa (atmospheric pressure effect); (2) Wind stress: $(V_ms/50)^2 \times 3$ m, normalised to a Cat-4 reference; (3) Distance decay: $(1 - d/500)^{0.5}$ for $d < 500$ km; (4) Approach factor: increases if the 48h track forecast passes closer to the station; (5) Station exposure weight (0.5–1.0 based on coastal geometry). Result is capped at 10 m and classified LOW/MODERATE/HIGH/EXTREME.

■ Why not a trained ML surge model?

High-quality labelled training data for BoB storm surge requires tide gauge records matched to historical cyclone landfall events — a sparse, difficult-to-obtain dataset. Numerical models (ADCIRC, FVCOM) require bathymetry and are computationally expensive. The heuristic is explicitly labelled as an approximation and is appropriate for a thesis prototype.

■ What are the main sources of error in the surge heuristic?

Bathymetric amplification (the Meghna estuary and Sundarbans funnel surge dramatically), storm size (Rmax not modelled), translation speed (slow storms produce more surge), and tidal interaction (surge + high tide >> surge alone). These are listed in Limitations.

7. Technical Deep Dives

Q1: How did you handle the 269 MB track model file that exceeds GitHub's 100 MB limit?

The trained model is excluded from the repository via .gitignore. Instead, the training script (scripts/train_track_predictor.py) is committed, along with the IBTrACS data CSV (ibtracs_bob.csv, 27 MB). Anyone cloning the repo runs `python scripts/train_track_predictor.py` to regenerate the model in ~3 minutes. The compressed model (lzma level 3) is 37 MB and can be shared separately via Telegram or similar.

Q2: Why Python 3.12 for V3 instead of 3.14?

PyTorch 2.2.2 does not provide compiled wheels for Python 3.14. Using pyenv, Python 3.12.3 was installed alongside the system Python, and V3 training was run with `~/.pyenv/versions/3.12.3/bin/python3`. This is a temporary workaround until the PyTorch project releases 3.14 wheels.

Q3: What is GroupShuffleSplit and why was it used?

GroupShuffleSplit is a scikit-learn cross-validation splitter that ensures all observations from a given group (here: storm SID) are entirely in either train or test — never both. This prevents the model from learning storm-specific patterns that inflate evaluation scores but would not generalise to new storms.

Q4: What is the difference between ROC-AUC and PR-AUC and when does each matter?

ROC-AUC plots true positive rate vs. false positive rate across all thresholds. It is insensitive to class balance, making it easy to compare across experiments. PR-AUC plots precision vs. recall, which penalises false positives heavily when positives are rare. For the 120–168h label (rare intensification events), PR-AUC provides a more honest picture. Both should be reported for this project.

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8. Presentation Tips

Q1: How should I present the V1→V2→V3 progression?

Frame it as a systematic ablation: each version adds exactly one new data modality. Show the AUC table (0.58 → 0.67 → 0.89 at 48–72h) and emphasise the 120–168h jump (0.10 → 0.80) as the most scientifically interesting finding — it shows that environmental context is essential for long-range prediction. This is the thesis's central contribution.

Q2: What if a professor asks why you didn't use ECMWF or GFS forecasts as features?

Acknowledge it as a valid improvement: ECMWF ensemble forecasts would provide future environmental state at the prediction horizon, unlike ERA5 which is sampled at the current time. This is listed under Future Work as 'steering wind fields'. The constraint was that ECMWF reanalysis (ERA5) is free but ECMWF forecasts require institutional access; GFS forecasts were not integrated due to time constraints.

Q3: What if asked about the small training set (21 storms)?

Acknowledge it directly: 21 storms with full ERA5 coverage (2018–2020) is a small sample. Mitigating factors: IBTrACS provides 6-hourly observations, giving ~400–500 labelled data points; XGBoost is well-suited to small datasets; results are validated on held-out 2020 storms. Future improvement: multi-basin training or extending ERA5 coverage to 2010–2024.

Q4: What is the 'honest' summary of the project's accuracy?

Intensification (48–72h): AUC 0.89 with satellite images (V3), 0.67 without (V2). Intensification (120–168h): AUC 0.80 (V2). Both evaluated on held-out 2020 storms. Track (24h): MAE ~18 km leave-one-storm-out CV — competitive but not operational. Track (48h): MAE ~134 km — useful for direction indication only. Surge: physics heuristic, not validated against tide gauge observations.